Procedural-Skill SFT Across Capacity Tiers: A W-Shaped Pre-SFT Trajectory and Regime-Asymmetric Mechanism on 0.8B–4B Qwen3.5 Models

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Abstract

We measure procedural-skill SFT contribution across three Qwen3.5 dense scales (0.8B, 2B, 4B) on a 200-task / 40-skill holdout, with Claude Haiku 4.5 as a frontier reference. The corpus is 353 rows of (task + procedural-skill block, Opus chain-of-thought, judge-pass) demonstrations.

Main finding. Under matched-path LLM-only scoring, the SFT-attributable procedural-Δ lift is roughly uniform across the three sizes: +0.070 / +0.040 / +0.075 at 0.8B / 2B / 4B. Variation in post-SFT Δ across sizes (−0.005, +0.100, +0.065) is dominated by a W-shaped pre-SFT base trajectory (−0.075, +0.060, −0.010, with frontier Haiku-4-5 at +0.030): the 5-step procedure hurts very small (0.8B) and competent-but-not-frontier (4B) bases, helps an intermediate (2B) base, and helps a frontier base modestly. SFT works hardest in absolute terms where the base struggles with the procedure — a regime-asymmetric pattern with a falsifiable prediction at 8B/14B.

Methodological observations. (i) A bench format-compliance artifact: 83.5% of the holdout uses a deterministic `ANSWER`-line extractor that under-counts free-form-prose conclusions. Our LLM-only re-judge closes the gap and reveals the deterministic judge had been systematically biased *against* the curated condition. (ii) A negative-iteration sequence at 0.8B: five recipe variants (corpus composition, partial fine-tuning, skill-block stripping) cluster post-SFT CURATED pass-rate within a 2 pp band, constraining the absolute-pass-rate ceiling to base capacity rather than recipe.

Cross-family judge validation. GPT-5.4 via OpenRouter on all 7 configurations (2800 paired episodes) agrees on the direction of every per-student finding: Cohen's $\kappa \geq 0.754$, agreement $\geq 93.25\%$, max headline Δ shift ≤ 0.035 pp.

Two earlier framings — "format-only learning at 0.8B" and "SFT contribution shrinks at 4B" — were path-mismatch artifacts; this paper supersedes both (Appendix A). Single-seed evaluation; threats itemised in §10.

1 Introduction

A common compositional-skills hypothesis holds that LLM capabilities can be decomposed into reusable skills [1, 2], and that explicit procedural instructions can elicit better task performance from smaller students than implicit-skill priming alone [3]. A natural follow-up: does demonstrating the explicit procedure during fine-tuning teach a smaller student to *use* such procedures more effectively at inference time, or merely to imitate the response shape?

We answer this empirically on a self-contained pipeline: a hand-curated 40-skill procedural catalog, an Opus-synthesized 200-task holdout, a 353-row Opus-traced curated SFT corpus, three Qwen3.5 LoRA students (0.8B/2B/4B), and a matched-path evaluation protocol with both deterministic and LLM-judge scoring. Throughout, Opus 4.7 was used for skill rewriting, task synthesis, baseline trace generation, and judge-of-record scoring; this judge-overlap is a structural threat to validity we address quantitatively via cross-family validation (§5) and itemise explicitly (§10).

Empirical contributions.

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1. **Cross-scale SFT contribution under matched-path scoring.** The SFT-attributable procedural-Δ lift is roughly uniform at +0.070 / +0.040 / +0.075 across 0.8B / 2B / 4B — a 4 pp band across an order of magnitude in base capacity (§3).
2. **W-shaped pre-SFT base trajectory and regime-asymmetric SFT mechanism.** Pre-SFT Δ is non-monotone with two negative pockets (0.8B and 4B). SFT works hardest where the base struggles with the procedure, and less where the base is already in a procedure-friendly regime — a falsifiable prediction at 8B/14B (§3.3).
3. **v2.0 ties the frontier reference at the bench's effective ceiling.** v2.0 CURATED = 0.985 matches Haiku-4-5 CURATED = 0.985 (197/200 identical pass count); pre-SFT 4B CURATED = 0.840 rules out the deflationary alternative that base scaling alone closes the gap (§3.5).

Methodological contributions.

4. **Bench format-compliance artifact.** The deterministic ANSWER-line extractor used by 83.5% of the bench under-counts free-form-prose conclusions, biasing similar SFT-on-procedural-demonstrations claims absent a format-tolerant base-model control. Our LLM-only re-judge reveals the bias was systematically against the curated condition (§4).
5. **Negative-result iteration sequence at 0.8B.** Five recipe variants cluster post-SFT CURATED within 2 pp regardless of corpus composition, partial fine-tuning, or skill-block stripping. The 0.8B absolute-pass-rate ceiling is base-capacity-bound, not recipe-bound (§6).

We release the pipeline, training/eval scripts, recipes, and full episode-level logs.

2 Pipeline and Bench

2.1 Skill catalog, procedural rewrite, and task synthesis

We hand-curate 40 declarative skills across categories (logical fallacies, deductive forms, cognitive biases, interpretive patterns, spatial reasoning), seeded from Skill-Mix Table 5/6 [1]. Opus 4.7 (1M context) rewrites each declarative entry into a procedural form following SkillsBench [3]: ordered procedure steps, when-to-use criteria, constraints, and a positive/negative worked example. A task synthesizer prompts Opus to produce 15 tasks per skill; the 600 generated tasks are split into 400 train / 200 eval, disjoint by task-uid with a 1:1 task-to-skill map. Tasks carry one of four query types: `YES_NO` (61%), `FREE_FORM` (16.5%), `SINGLE_WORD` (11.5%), `RANKING` (11%). Each task is generated knowing its target skill; the bench therefore measures procedural application on procedure-aligned tasks, not skill-routing or compositional skill use (§10).

2.2 SFT corpus and evaluation protocol

Each training task is traced by Opus under a SOLVER system prompt with the corresponding procedural skill block injected. Episodes that pass the LLM-judge [7] are filtered into a 353-row curated-only chat-format corpus: `(system: SOLVER + skill block, user: task, assistant: CoT + ANSWER line)`.

For evaluation, we generate one student response per `(task, condition)` pair on the 200-task hold-out. Two conditions: BASELINE (SOLVER prompt without skill block) and CURATED (SOLVER prompt with the matching procedural skill block). Decoding: greedy (T=0), `repetition_penalty`= 1.05, `max_new_tokens` 1024 post-SFT / 2048 pre-SFT.

Judge dispatch has two paths. *Deterministic-mixed* (the bench's native dispatch): deterministic `ANSWER`-line extraction for `YES_NO`/`SINGLE_WORD`/`RANKING`, Opus 4.7 LLM-judge for `FREE_FORM`. *LLM-only*: Opus 4.7 LLM-judge for every task type, used in §4 to bypass the format-compliance gate. We report pass-rate (binary verdict) and Δ = pass_rate(CURATED) − pass_rate(BASELINE). All numbers are single-seed.

2.3 Training

LoRA [4] via TRL $\geq$ 0.18 [6] and PEFT $\geq$ 0.13. Adapters target all-linear modules. The chat template is patched with `{% generation %}...{% endgeneration %}` markers to enable assistant-only loss masking. For partial-FT variants (v1.7, v1.8) we freeze all but the top-N transformer layers + `lm_head` + final norm, using bitsandbytes 8-bit Adam [5]. Base models are the Qwen3.5 small dense series [10], which builds on the Qwen3 architecture [9].

3 Cross-Scale SFT Contribution and the W-Shape

We measure the SFT contribution at each model scale by comparing each post-SFT student against its matched-path pre-SFT control under LLM-only scoring. Table 1 reports the contribution decomposition; Table 2 gives the underlying pass rates for all seven configurations.

	Pre-SFT base (LLM-only)			**SFT contribution**		
	BL	CU	Δ	ΔBL	ΔCU	Δ-lift
0.8B (v1)	0.665	0.590	−0.075	+0.045	+0.115	**+0.070**
2B (v1.9)	0.755	0.815	+0.060	+0.060	+0.100	**+0.040**
4B (v2.0)	0.850	0.840	−0.010	+0.070	+0.145	**+0.075**

Table 1: SFT contribution across all three model sizes under matched HF + LLM-only scoring. The Δ-lift attributable to SFT is in a tight 4 pp band (+0.040 to +0.075); the absolute CURATED contribution is in a similar band (+0.100 to +0.145). Variation in post-SFT Δ across sizes is dominated by the pre-SFT base trajectory, which is W-shaped (negative at 0.8B and 4B, positive at 2B and frontier Haiku-4-5; see Table 2).

	BL	**CU**	Δ	**McNemar** p	**bootstrap 95% CI on** Δ
Pre-SFT base controls					
pre-SFT 0.8B (HF)	**0.665**	**0.590**	**−0.075**	—	—
pre-SFT 2B (HF)	0.755	0.815	+0.060	—	—
pre-SFT 4B (HF)	**0.850**	**0.840**	**−0.010**	—	—
pre-SFT Haiku-4-5 (ref)	0.955	0.985	+0.030	—	—
Post-SFT students					
v1 (0.8B+LoRA)	0.710	0.705	−0.005	—	—
v1.9 (2B+LoRA)	0.815	0.915	+0.100	0.0027	[+0.040, +0.160]
v2.0 (4B+LoRA)	0.920	0.985	+0.065	0.00098	[+0.030, +0.105]

Table 2: Pass rates for all seven configurations under matched HF + LLM-only scoring. Haiku-4-5 was evaluated via the Anthropic API; all Qwen3.5 configurations were evaluated via HF transformers locally. Three load-bearing observations: (i) v2.0 CURATED ties the Haiku reference at 0.985 (197/200 identical pass count); (ii) pre-SFT base Δ is W-shaped — $-0.075, +0.060, -0.010, +0.030$ — with negative pockets at 0.8B and 4B; (iii) the SFT contribution to Δ-lift is roughly uniform across all three sizes (Table 1).

3.1 The SFT contribution is roughly capacity-invariant in differential terms

Across the three model sizes under matched-path LLM-only scoring, the SFT-attributable Δ-lift sits in a 4 pp band: +0.070 at 0.8B, +0.040 at 2B, +0.075 at 4B. The absolute CURATED contribution is in a similar band: +0.115 at 0.8B, +0.100 at 2B, +0.145 at 4B. Per-model McNemar tests under LLM-only judging confirm v1.9 ($\chi^2 = 9.03$, $p = 0.0027$, CI $[+0.040, +0.160]$) and v2.0 (exact, $p = 0.00098$, CI $[+0.030, +0.105]$) each have a positive Δ with bootstrap CI excluding zero (Table 7). The 0.8B SFT Δ-lift is judge-invariant within 1 pp: +0.065 under deterministic-mixed judging, +0.070 under LLM-only (§4 explains the per-judge gap).

The differential component is what distinguishes "the model uses the injected procedure" from "the model just imitates the response shape." Pure format-learning would lift BASELINE and CURATED equally and leave Δ untouched, which we do not see at any of the three scales — Δ-lifts are +0.070, +0.040, +0.075. We resist a quantitative "half format, half procedure" decomposition: the format and procedure components are not independent (producing the procedural-step shape implicitly enforces a checking discipline that may itself be the procedural lift), and the underlying contributions cannot be cleanly separated by per-condition pass-rates alone. The differential Δ-lift is evidence that a procedure-vs-no-procedure component of the gain exists; it is not a quantitative apportionment.

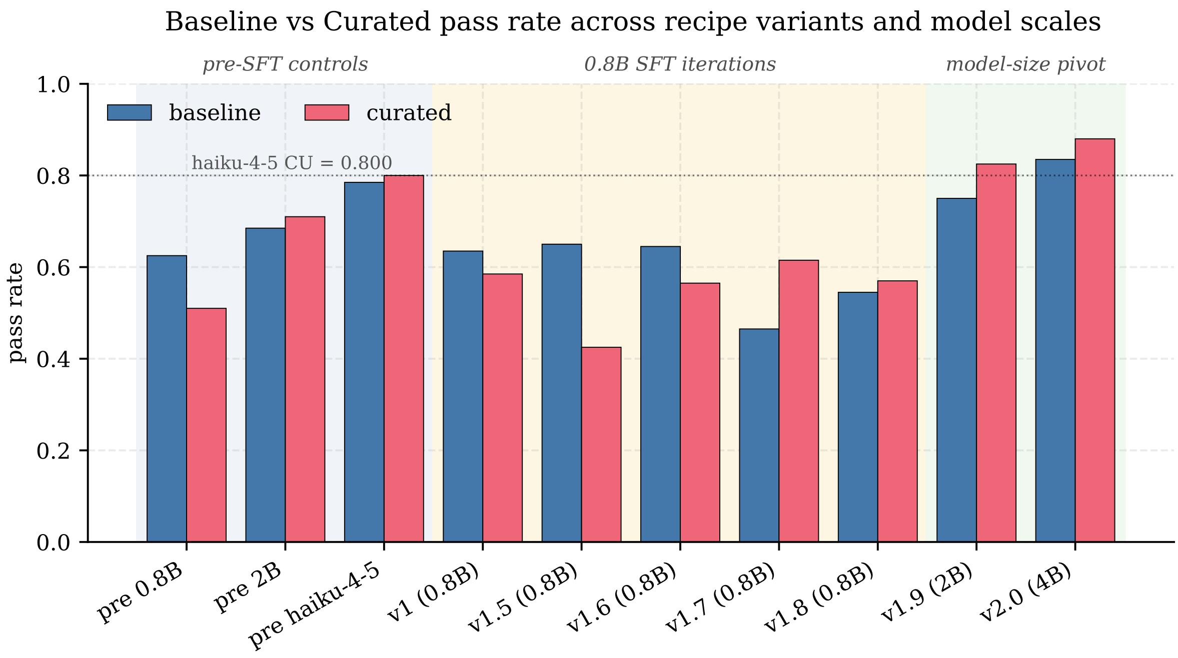

Figure 1: Baseline (BASELINE) and curated (CURATED) pass rates across all eleven evaluated configurations: pre-SFT controls, the five 0.8B SFT iterations (§6), and the model-size pivot to 2B and 4B.

3.2 W-shaped pre-SFT base trajectory

The pre-SFT base trajectory of procedural responsiveness is non-monotone with two negative pockets:

- **pre-SFT 0.8B (HF, LLM-only):** $\Delta = -0.075$ — **deepest trough**
- pre-SFT 2B (HF, LLM-only): $\Delta = +0.060$
- **pre-SFT 4B (HF, LLM-only):** $\Delta = -0.010$ — **shallow trough**
- pre-SFT Haiku-4-5 (LLM-only): $\Delta = +0.030$

Both 0.8B and 4B are in regimes where the procedure injection actively hurts the base; 2B and Haiku are in regimes where it helps. The two troughs likely reflect different mechanisms:

- *0.8B (deep trough):* the base lacks the capacity to follow the 5-step procedure correctly. Errors compound at every step; the procedure structure adds sub-inference error opportunities the base cannot recover from.
- *4B (shallow trough):* the base is competent enough to solve baseline tasks directly. The procedure adds overhead without adding signal — curated responses sometimes follow the procedure into less-defensible conclusions on tasks the base would have one-shotted.

Frontier-class Haiku partially recovers by having enough native capacity to use the procedure as intended without being captive to it. 2B sits in a sweet spot where the procedure scaffolds checking the model would otherwise skip.

3.3 Regime-asymmetric SFT mechanism

The roughly capacity-invariant Δ-lift magnitude is doing visibly different work at different scales:

- **0.8B** (pre-SFT $\Delta - 0.075$, post-SFT $\Delta - 0.005$): SFT lifts by $+0.070$, rescuing a deeply procedure-incompetent regime to procedure-neutral. SFT teaches the model to track the procedure without the compounding errors.
- **2B** (pre-SFT $\Delta + 0.060$, post-SFT $\Delta + 0.100$): SFT lifts by $+0.040$, marginally improving an already-procedure-friendly regime. The base is in the sweet spot; SFT adds discipline but has less headroom.
- **4B** (pre-SFT $\Delta - 0.010$, post-SFT $\Delta + 0.065$): SFT lifts by $+0.075$, flipping a procedure-overhead regime into procedure-helps. SFT teaches the model to use the procedure productively rather than treat it as friction.

The pattern: SFT works hardest (in absolute Δ-lift terms) where the base struggles with the procedure, and less where the base is already in a procedure-friendly regime. The "uniform" framing is fair as a magnitude bound; the underlying mechanism is regime-asymmetric.

Falsifiable prediction at 8B/14B. If the mechanism explanation is correct, at 8B/14B the procedure should hurt the base less than at 4B (since the base can use it natively, like Haiku), and SFT's Δ-lift should shrink toward 2B-ish levels. If pre-SFT Δ recovers toward Haiku-class positive Δ as base capacity grows, and SFT's Δ-lift correspondingly shrinks in the procedure-friendly regimes, the W-shape and regime-asymmetric story extend. If pre-SFT Δ stays negative or SFT's Δ-lift stays uniform, the mechanism explanation needs revision.

3.4 Per-skill evidence supports the mechanism

The regime-asymmetric pattern recurs at the skill level. Figure 2 shows the v2.0 procedural-Δ broken down by skill (deterministic-judge counts). Eleven skills lift on CURATED (concentrated on interpretive and spatial-social skills); 25 are flat (most because BASELINE is at ceiling and CURATED has nowhere to go); 4 regress under deterministic judging (deductive-logic skills where the base has ceiling competence, mirroring the 4B pre-SFT trough at the skill level). Under LLM-only judging the regression cluster shrinks to one skill.

Figure 3 plots per-skill Δ against per-skill BASELINE at v2.0 under LLM-only judging. Spearman $\rho = -0.227$ across n=40 skills (two-sided $p \approx 0.15$, underpowered: 29/40 skills are tied at BASELINE$= 1.0$, collapsing to the same rank). On the non-ceiling subset (skills with BASELINE < 1.0, n=11), the correlation strengthens to $\rho = -0.455$ ($p \approx 0.13$); the negative slope is qualitatively consistent in both subsets but neither reaches conventional significance at this n. The lift cluster has mean baseline 0.71; the flat cluster has mean baseline 1.00. We treat this as qualitative evidence rather than a load-bearing statistical claim: the procedure helps where the base struggles and is silent where the base is at ceiling — the per-skill miniature of the cross-scale regime-asymmetric finding.

3.5 v2.0 ties Haiku at the bench's effective ceiling

Under fair scoring, v2.0 CURATED = 0.985 matches the Haiku-4-5 reference at CURATED = 0.985 (197/200 identical pass count). With pre-SFT 4B measured at CURATED = 0.840 under fair judging, the v2.0=Haiku tie is attributable to SFT plus base scaling: the 4B base does *not* already match Haiku before SFT (SFT contributed +0.145 CURATED on top of base 4B). An earlier framing where "2B+SFT crosses Haiku" under the deterministic-mixed dispatch (0.825 vs 0.800) was an artifact of the format-compliance gate (§4): Haiku's free-form prose was disproportionately rejected by the `ANSWER`-line extractor, and under LLM-only judging Haiku's CURATED rises by +0.185 to the ceiling — more than any other student. Under LLM-only judging, v1.9 CURATED (0.915) does *not* cross Haiku; only v2.0 does.

We resist a sharp compute-efficiency claim relative to Haiku: its parameter count is unpublished (community estimates 10^{10}–$3{\times}10^{10}$ parameters, no anchor); Opus 4.7 wrote the corpus content, generated the SFT traces, and judged both students — a structurally biased judging path (§5 partially closes this); and the SFT student is in-distribution for the bench while Haiku is asked cold. The honest summary: under fair judging, v2.0 (4B+SFT) and Haiku tie at the bench's curated ceiling on tasks of this construction style.

3.6 The bench saturation diagnosis splits in two

The v1.9→v2.0 post-SFT Δ shrinkage (+0.100 → +0.065 under LLM-only) is directionally consistent with bench saturation but not statistically distinguishable from sampling noise on the v1.9-vs-v2.0 adjacent-pair test: one-sided bootstrap $p = 0.191$ under LLM-only judging ($p = 0.225$ under deterministic-mixed), 95% CI on $\Delta_{\text{v1.9}} - \Delta_{\text{v2.0}}$ is $[-0.040, +0.105]$, which includes zero. Saturation has two components, and they diverge:

- *Absolute pass rates:* saturated. v2.0 CURATED = 0.985 ties Haiku at the bench's effective ceiling.
- *SFT mechanism:* not saturated. Δ-lift is roughly uniform across all three sizes (+0.040 to +0.075), and grows from 2B's +0.040 to 4B's +0.075 rather than shrinking.

What changes across the post-SFT trajectory is the pre-SFT starting point (W-shape), not the SFT mechanism's strength. An earlier version of this paper conflated absolute saturation with mechanism saturation under one label; only the absolute reading survives.

4 Format-Compliance Artifact and LLM-Only Remediation

4.1 The artifact

83.5% of the bench (167/200 tasks) uses a deterministic `ANSWER`-line extractor for scoring. Base Qwen3.5-4B reasons correctly on bench tasks (spot-checks confirm step-by-step reasoning that reaches the correct answer) but does not consistently end with `ANSWER: <X>`, so the deterministic dispatch returns failure. The bench's 0/200 for pre-SFT 4B under deterministic-mixed scoring should be read as "format-compliance baseline ≈ 0," not "reasoning ≈ 0."

This generalises: any post-SFT lift measured against a format-strict deterministic judge conflates three contributions — format learning (the model emits the required line), format-conditional reasoning gating (the model reasons more reliably when constrained to structured output), and procedural application (the model uses the injected skill). Without a base-model control evaluated under format-tolerant scoring, these are not separable. Plausible remedies: (a) a stricter base-model evaluation prompt that reliably hits the format target, (b) a format-tolerant judge for base controls (e.g. extracting the answer with an LLM call rather than a regex), (c) restricting evaluation to `FREE_FORM` tasks. We apply remedy (b).

4.2 LLM-only re-judge: closing the format gap

We re-route every episode through the LLM judge regardless of task type. For the post-SFT students and Haiku we re-ran the LLM judge over existing model responses (no regeneration); for the pre-SFT base models we ran fresh HF-transformers evaluations with `-force-llm-judge`. Numbers are in Table 2; Table 3 shows the score shifts from deterministic-mixed.

	ΔBL	ΔCU	$\Delta\Delta$
v1 (0.8B+LoRA)	+0.075	+0.120	+0.045
v1.9 (2B+LoRA)	+0.065	+0.090	+0.025
v2.0 (4B+LoRA)	+0.085	+0.105	+0.020
Haiku-4-5 (ref)	+0.170	+0.185	+0.015

Table 3: Score shifts from deterministic-mixed to LLM-only re-judge. All four students gain on both conditions; Haiku gains the most. For all three SFT'd students, ΔCU > ΔBL by 2–4.5 pp, meaning the deterministic judge was systematically biased *against* the curated condition — procedure-following responses bury conclusions in narration more often than direct-thinking responses do.

All pass rates rise; Haiku rises the most. The deterministic-mixed judge under-counted passes universally. Even the SFT'd students, trained to produce `ANSWER` format, occasionally emitted alternative phrasings ("Final Answer: X," "I conclude X," or stating the answer in prose) that the regex missed. Haiku, never trained on the bench's response format, was penalised the most — +17 pp on BASELINE and +18.5 pp on CURATED. This is the cleanest direct evidence that the format-compliance gate biased the bench against students whose response style does not match the SFT distribution.

The deterministic judge was biased against the curated condition. For all three SFT'd students, ΔCU exceeds ΔBL by 2–4.5 pp under re-judge (Table 3). When the procedure block is present, the model occasionally embeds its conclusion inside procedural narration ("...therefore the conclusion follows: X") rather than ending with a literal `ANSWER: X` line. The deterministic judge under-counted curated wins systematically, under-stating the SFT contribution. Under LLM-only judging, v2.0's p=0.00098 comfortably survives Bonferroni correction across the two per-model tests; the deterministic-mixed p=0.049 would not.

0.8B path-mismatch implication. The original pre-SFT 0.8B baseline was run via Ollama on a different code path than the v1 post-SFT evaluation, producing artificially low baseline numbers and an apparent "Δ-flip" that earlier versions of this paper read as "format-only learning at 0.8B." Matched-path

HF + LLM-only scoring resolves the mismatch and reveals the SFT Δ-lift at 0.8B is +0.070 (LLM-only) or +0.065 (deterministic-mixed), on par with 4B. The full reconciliation, including the methodology for the matched-path deterministic-mixed pre-SFT 0.8B number, is in Appendix A.

5 Cross-Family Judge Validation

The judge-overlap concern (Opus 4.7 used for skill rewriting, task synthesis, SFT-trace generation, and judging) is the strongest single-axis methodological threat in the experiment. To bound the magnitude, we ran a non-Anthropic-family second judge (GPT-5.4 via OpenRouter) on the LLM-only re-judge episodes for all seven configurations (Table 4).

	Opus 4.7 LLM-only		**GPT-5.4 (OpenRouter)**		
	BL / CU	Δ	BL / CU	Δ	Δ shift
v1 (0.8B+LoRA)	0.710 / 0.705	−0.005	0.700 / 0.700	0.000	+0.005
v1.9 (2B+LoRA)	0.815 / 0.915	+0.100	0.815 / 0.915	+0.100	0.000
v2.0 (4B+LoRA)	0.920 / 0.985	+0.065	0.920 / 0.980	+0.060	−0.005
Haiku-4-5 (ref)	0.955 / 0.985	+0.030	0.955 / 0.975	+0.020	−0.010
pre-SFT 0.8B	**0.665 / 0.590**	**−0.075**	**0.670 / 0.585**	**−0.085**	−0.010
pre-SFT 2B	0.755 / 0.815	+0.060	0.750 / 0.835	+0.085	+0.025
pre-SFT 4B	**0.850 / 0.840**	**−0.010**	**0.850 / 0.805**	**−0.045**	**−0.035**

Table 4: Cross-family judge cross-validation across all seven configurations / 2800 paired episodes. Per-episode agreement ranges from 93.25% (pre-SFT 4B) to 100.00% (v1.9, 400/400 identical verdicts). Cohen's κ ranges from 0.754 to 1.000. Pre-SFT 0.8B HF agreement is 98.50% ($\kappa = 0.968$). Headline Δ shifts ≤ 0.035 pp on every dataset; both judges agree on the direction of every per-student finding, including the W-shape's two negative-Δ pockets at 0.8B and 4B.

Across 2800 paired episodes, the maximum headline-Δ shift between judges is 0.035 pp (pre-SFT 4B); the minimum agreement is 93.25%; the minimum κ is 0.754. Under a non-Anthropic-family second judge on non-Anthropic infrastructure, no headline conclusion changes direction across any of the seven configurations. Both negative-Δ pockets in the W-shape are judge-invariant: pre-SFT 0.8B has $\Delta \leq -0.075$ (Opus −0.075, GPT −0.085), pre-SFT 4B has $\Delta \leq 0$ (Opus −0.010, GPT −0.045). Cross-family bounds on the SFT contribution: at 0.8B, Δ-lift $[+0.070, +0.085]$; at 2B, Δ-lift $[+0.015, +0.040]$ (GPT credits base 2B with more procedural responsiveness than Opus does); at 4B, Δ-lift $[+0.075, +0.105]$. All three SFT contributions are positive under both judges.

What this validation does not close. It addresses only the *judging-stage* overlap. Opus still wrote the procedural rewrites, synthesized the tasks, and generated the SFT-corpus traces; this corpus-generation overlap is structurally unbounded by judging-stage cross-validation. A non-Anthropic third-family judge (Gemini, Llama-3.3-70B) or human raters would extend the bound; we did not run them. The most plausible remaining direction of bias is in the corpus-generation stage rather than the judging stage.

6 Recipe Iteration at 0.8B

Before scaling to larger bases, we ran five LoRA-recipe variants at 0.8B (v1–v1.8) to test whether the small-base bottleneck was in the corpus, the optimizer, or the chat-template handling. Table 5 reports all eleven configurations under the bench's native deterministic-mixed dispatch.

6.1 Recipe variation does not move the 0.8B ceiling

Three well-formed 0.8B variants (v1, v1.6, v1.8) cluster CURATED within a 2 pp band (0.565–0.585 deterministic, $\sim$ 0.705 for v1 under LLM-only re-judge). The cluster does not move with corpus composition (v1.5 adds BASELINE +CURATED mixed rows; collapses CURATED to 0.425), ceiling-skill dropping (v1.6 removes skills where the base passes $\geq$ 0.90; lands within v1's band), or skill-block stripping (v1.8 trains with the SKILL block removed from system prompts; lands within v1's band). The post-SFT absolute CURATED pass rate at 0.8B is bounded by base 0.8B's reasoning capability, regardless of recipe.

Variant	Base	Recipe	BL	CU	Δ
Pre-SFT 0.8B	0.8B (HF)	none	0.625	0.510	−0.115 §
Pre-SFT 2B	2B (HF)	none	0.685	0.710	+0.025
Pre-SFT 4B	4B (HF)	none	0.000 †	0.000 †	—
Pre-SFT Haiku-4-5 (ref)	claude-haiku-4-5	none	0.785	0.800	+0.015
v1	0.8B	LoRA r=8, curated-only, 353 rows	0.635	0.585	−0.050
v1.5	0.8B	LoRA + BL+CU mixed corpus	0.650	0.425	−0.225
v1.6	0.8B	LoRA + drop ceiling skills + chat-patch	0.645	0.565	−0.080
v1.7	0.8B	partial FT (top-6 layers + lm_head)	0.465	0.615	+0.150 ‡
v1.8	0.8B	partial FT + skill-block stripped from training	0.545	0.570	+0.025
v1.9	2B	LoRA r=16, curated-only, 353 rows	**0.750**	**0.825**	**+0.075**
v2.0	4B	LoRA r=32, curated-only, 353 rows	**0.835**	**0.880**	**+0.045**

Table 5: Recipe-by-recipe pass rates on the 200-task holdout under deterministic-mixed scoring (the bench's native dispatch). All bases are Qwen3.5 dense models except where noted. All evaluations use HF transformers generation; the original Ollama-path pre-SFT 0.8B numbers (0.510/0.565/ + 0.055) are documented in Appendix A. † Pre-SFT 4B's 0/0 is a format-compliance artifact (§4). ‡ v1.7's apparent +Δ is an artifact of BASELINE collapse, not differential learning. § The matched-path methodology for pre-SFT 0.8B is in Appendix A. LLM-only re-judge values (Table 2) reframe the SFT attribution at 0.8B; Δ-lift is +0.070 matched-path LLM-only, +0.065 deterministic-mixed.

v1.7's apparent Δ is a mode-collapse artifact. Replacing LoRA with full-rank partial fine-tuning of the top-6 layers + `lm_head` + final `norm` (using 8-bit Adam to fit consumer GPU budget) produces an apparent Δ = +0.150, but with BASELINE *collapsed* to 0.465 — substantially below the matched-path pre-SFT 0.8B BASELINE of 0.625. Inspection of v1.7 BASELINE failures reveals the model emits "`Step 1 (SKILL block: ...)`" prefaces and procedural step language even when no skill block is in the system prompt: the model has learned to expect a skill block and hallucinates one when absent. The +0.150 Δ does not reflect the model learning to use skill blocks; it reflects the model losing the ability to function without one. v1.8 (skill-block stripped from training) recovers BASELINE to 0.545 and lands CURATED back inside the 0.565–0.585 band, confirming the diagnosis.

What the 0.8B negative results constrain, and what they do not. What clusters at 0.8B is the post-SFT *absolute* pass rate, which is base-capacity-bound. What does *not* cluster is the SFT-attributable Δ-lift, which is +0.070 at 0.8B under matched-path scoring — comparable to the 4B Δ-lift of +0.075 (Table 1). The capacity-floor diagnosis was correct for absolute pass rate; under earlier path-mismatched scoring it was wrongly extended to the SFT mechanism. Under matched-path scoring, SFT delivers comparable differential work at 0.8B as at 4B; what it cannot do at 0.8B is push the absolute ceiling.

6.2 v1.9 (2B) attribution split

Same recipe shape as v1 (LoRA, curated-only, 353 rows, 3 epochs), bumped to r=16/α=32 to give the larger base more LoRA capacity, on Qwen3.5-2B. Result: BASELINE 0.750 / CURATED 0.825 / Δ +0.075 (paired-bootstrap 95% CI [+0.020, +0.130]; McNemar χ^2=5.60, p=0.018).

7 Statistical Tests

Each task is run in both BASELINE and CURATED conditions, so verdicts are paired. We run McNemar's test (continuity-corrected χ^2 for $n_{\text{discordant}} \geq 25$, exact binomial otherwise) per model, and a paired bootstrap (10,000 resamples of `task_uid` with replacement) for 95% CIs on Δ. For the cross-model saturation test we resample v1.9 and v2.0 independently and report a one-sided bootstrap test for $H_0 : \Delta_{\text{v1.9}} \leq \Delta_{\text{v2.0}}$.

	deterministic-mixed		LLM-only (matched-path)	
	ΔBL	ΔCU	ΔBL	ΔCU
Base scaling 0.8B → 2B	+0.060	+0.200	+0.090	+0.225
SFT contribution at 2B	+0.065	**+0.115**	+0.060	**+0.100**
Total v1.9 vs pre-SFT 0.8B	+0.125	+0.315	+0.150	+0.325
SFT Δ-lift at 2B	+0.025 → +0.075 (**+0.050**)		+0.060 → +0.100 (**+0.040**)	

Table 6: v1.9 attribution split under both judging paths. The deterministic-mixed columns use matched-path pre-SFT 0.8B (HF, 0.625/0.510; Appendix A) and pre-SFT 2B (HF, 0.685/0.710); the LLM-only columns use the corresponding LLM-only-judged controls. The SFT-attributable Δ-lift at 2B is +0.050 under deterministic-mixed and +0.040 under LLM-only — consistent within 1 pp across both judging paths even as absolute magnitudes shift.

	deterministic-mixed		LLM-only re-judge	
	Δ	McNemar p / 95% CI	Δ	McNemar p / 95% CI
v1.9	+0.075	χ^2=5.60, p=0.018 CI [+0.020, +0.130]	+0.100	χ^2=9.03, p=0.0027 CI [+0.040, +0.160]
v2.0	+0.045	exact, p=0.049 CI [+0.005, +0.085]	+0.065	exact, p=0.00098 CI [+0.030, +0.105]

Table 7: Per-model paired tests on the procedural-Δ, under both judging paths. Both models have a positive Δ with bootstrap CI excluding zero in both paths. Under LLM-only re-judge, v2.0's p=0.00098 comfortably survives Bonferroni correction across the two tests (the deterministic-mixed p=0.049 would not). Neither test is replicated across seeds.

For the cross-model saturation test, the result is essentially unchanged across judging paths: deterministic-mixed $\Delta_{\text{v1.9}} - \Delta_{\text{v2.0}} = +0.030$, CI [−0.040, +0.100], one-sided $p = 0.225$; LLM-only $\Delta_{\text{v1.9}} - \Delta_{\text{v2.0}} = +0.035$, CI [−0.040, +0.105], one-sided $p = 0.191$. The CI includes zero in both paths. The Δ-shrinkage v1.9→v2.0 is consistent with bench saturation in direction but is not statistically distinguishable from sampling noise at n=200. The cross-scale SFT contribution growth (Δ-lift +0.040 → +0.075 from 2B to 4B; §3) is robust to the same noise because it is computed against a different pre-SFT baseline (pre-SFT 4B Δ −0.010).

8 Generality Probe

We probe v1.9, v2.0, base 2B, and base 4B with a 52-prompt OOD battery (10 generality + 42 factual including 8 false-premise "trick" prompts) under a generic system prompt (not the SOLVER prompt; Table 8). The probe is a convenience sample, not a calibrated benchmark; we treat it as qualitative evidence of OOD behavior.

	format-locked	factual HIT	trick CONFAB (eyeballed)	phantom skill block
v1.9 (2B)	0/52	33/34 †	6/8	0/52
base 4B	0/52	34/34	∼1/8	0/52
v2.0 (4B)	0/52	34/34	**0/8**	0/52

Table 8: Generality probe under generic system prompt. † v1.9 swapped *Brave New World* attribution from Aldous Huxley to H. G. Wells — a plausible-author confabulation absent from base 4B and v2.0.

v1.9 introduced a small calibration regression on adversarial prompts: the SFT-induced "produce a confident step-by-step answer" prior leaks into out-of-distribution behavior, making v1.9 commit to plausible-sounding inventions where base 2B would refuse. v2.0 does not exhibit this — the 4B base has enough native capacity to recognize false premises and refuse cleanly without the SFT shape overriding refusal. The bench-time finding that 17.5% of v2.0's BASELINE responses reference a "skill block" not present in the system prompt does *not* extend to general-chat distributions: under a generic system prompt, the SFT-induced skill-block reflex is dormant (0/52). The phantom-mention is contextual to

SFT-distribution-similar prompts.

9 Discussion

9.1 Mode collapse reinterpreted

The v1.7 mode collapse (partial-FT producing apparent $+\Delta$ at the cost of BASELINE integrity; §6) appeared at 0.8B as a structural concern about SFT-on-procedural-demonstrations and motivated the v1.8 skill-block-stripping intervention. Spot-checks of v2.0 BASELINE responses show the same phantom skill-block reference (35/200, 17.5%) but with 94.3% of those responses still passing: at 4B, the phantom mention is cosmetic rather than load-bearing. The model has the capacity to execute the procedure regardless of whether it remembers seeing a skill block in the system prompt. The mode collapse was a base-capacity-floor artifact at 0.8B, not a structural property of the SFT corpus. We do not claim the v1.8 skill-strip fix is unnecessary in general; we observe only that, at the model size where SFT produces a measurable differential lift in this study (4B), the failure mode it was designed to prevent is not load-bearing.

9.2 Negative results were informative

The five 0.8B variants are individually informative as negative results: corpus composition, partial fine-tuning, and chat-template patching all failed to lift the absolute pass rate at this scale and corpus size. They are jointly informative: together they constrain the diagnosis to "base-model capacity is the binding constraint on absolute pass rate at this corpus size," which the 2B run validates. Running through the negative-result series before scaling was epistemically efficient (∼\$15 in compute, three days of GPU time) compared to skipping straight to 4B with no priors on what fails. We argue iterated negative-result reporting is undervalued in the SFT-on-demonstrations literature: most claims in this space are backed by single-recipe positive results without surrounding falsification of nearby variants.

9.3 What we did not separately decompose

The differential Δ-lift is evidence that there is a procedure-vs-no-procedure component of the gain that pure format-learning cannot explain. We do not attempt a quantitative "x% format, y% procedure" decomposition: the format and procedure components are not independent (producing the procedural-step shape implicitly enforces a checking discipline that may itself be the procedural lift), and per-condition pass-rates do not separate them cleanly. A natural further ablation — training the same Qwen3.5 base on 353 rows of generic Opus instruction-tuning data and re-running the bench — would sharpen the specificity claim; we did not run it (§10).

10 Threats to Validity and Limitations

1. **Single-seed evaluation.** All headline numbers (Tables 2, 1, 7) are single-seed point estimates with within-seed paired statistics. The v2.0 deterministic McNemar p=0.049 should be read as "one trial that just barely cleared the conventional threshold," not as established significance; the LLM-only p=0.00098 is more robust but is also a single seed. 3–5 seed replication of v1.9 and v2.0 would bound run-to-run variance [8]; we did not run it.

2. **Corpus-generation overlap with the judge.** Opus 4.7 was used in four roles: rewriting the declarative skill catalog into procedural form, synthesizing the held-out tasks, generating the SFT-corpus traces, and judging both pre-SFT and post-SFT responses. The cross-family validation (§5, 2800 paired episodes, max headline Δ shift ≤ 0.035 pp, $\kappa \geq 0.754$) bounds the *judging-stage* component. The corpus-generation overlap is structurally unbounded by judging-stage cross-validation. A non-Anthropic third-family judge (Gemini, Llama-3.3-70B) or human raters would extend the bound to two non-Anthropic families; estimated cost ∼\$5–10 (LLM judge) or ∼\$50–100 (human raters) on a 200-episode subsample. The most plausible remaining direction of bias is in the corpus-generation stage rather than the judging stage.

3. **Single-skill-per-task synthesis.** Tasks were generated by Opus from a known skill, then the same skill was injected at evaluation time. The bench measures *procedural application on procedure-*

aligned tasks, not skill-routing or compositional skill use. This is closer to the Skill-Mix compositional hypothesis [1] than to a tautology — the student does have to follow the procedure correctly, and many tasks include constructed traps for procedure-skipping models — but it is narrower than "procedural skill use" as it appears in some recent work. The honest claim is that SFT'd students apply named procedures more effectively when shown a procedure aligned to a procedure-aligned task.

4. **Single model family (Qwen3.5).** All three SFT'd students and the two HF-evaluated pre-SFT controls are Qwen3.5 dense models. The W-shape we observe in pre-SFT Δ may be a Qwen3.5-specific phenomenon or a property of small-to-mid-scale dense LLMs in general; this paper cannot distinguish. A cross-architecture replication at a single capacity tier (e.g. Llama-3.2-3B as a 4B-class comparison) would be cheap and would substantially strengthen the architectural-generality claim. The Haiku-4-5 reference contributes one architecturally distinct data point but is not size-matched.

5. **No generic instruction-tuning control.** The paper's central distinction is between "the model uses the injected procedure" and "the model just imitates the response shape." Differential Δ-lift is the load-bearing evidence for the first reading. The specificity claim — that *procedural-skill* SFT delivers the +0.04 to +0.075 Δ-lift, as opposed to any 353-row Opus-trace SFT — is unwitnessed by the most natural ablation: training the same Qwen3.5 base on 353 rows of generic Opus instruction-tuning data and re-running the bench. Three outcomes the present data does not rule out: (a) generic SFT gives the same Δ-lift (the recipe teaches CoT discipline, not procedural specificity); (b) generic SFT gives smaller or negative Δ-lift (procedural specificity is real); (c) generic SFT gives a different cross-scale trajectory. Estimated cost: one additional training run per scale, $\sim$1 GPU-day total.

6. n=200 **per condition limits Δ-statistic resolution.** Per-condition binomial standard error is $\sim$3 pp. The cross-model bootstrap test for the v1.9-vs-v2.0 saturation hypothesis fails (p=0.191 LLM-only, p=0.225 deterministic). This only addresses the post-SFT Δ-shrinkage; the SFT-contribution comparisons across scales are robust to the same noise because they are computed from distinct pre-SFT baselines.

7. **Decoding is greedy throughout.** We did not sweep temperature or top-p; whether the post-SFT Δ is robust to sampling stochasticity is untested.

8. **Generality probe is a convenience sample.** The 52-prompt OOD battery (Table 8) is hand-written; trick-prompt CONFAB judgments were eyeballed. The qualitative claim "v2.0 preserves base 4B's out-of-distribution behavior" is supported by the data but should not be read as quantitative parity.

9. **Haiku inference path mismatch (minor).** Haiku-4-5 was evaluated through the Anthropic API while Qwen3.5 students used HF transformers locally. Inference parameters were matched as closely as possible; tokenization and decoding stack still differ. The load-bearing comparisons in this paper are within-Qwen3.5 across capacity tier, not Qwen3.5-vs-Haiku.

What this paper does not claim. We do not claim that pre-SFT 4B is below Haiku on general capability; only that, on this bench with this corpus, v2.0 (4B+SFT) and Haiku tie at the curated ceiling. We do not claim the W-shape generalises to architectures we did not test; we offer a falsifiable prediction at 8B/14B and at other architectures. We do not claim a quantitative format-vs-procedure decomposition; only that the differential Δ-lift evidences a procedure-vs-no-procedure component pure format-learning cannot explain.

11 Conclusion

Across three Qwen3.5 dense scales (0.8B, 2B, 4B) under matched-path scoring, procedural-skill SFT on a 353-row corpus delivers a roughly uniform Δ-lift in the +0.040 to +0.075 band. Variation in post-SFT Δ across sizes is dominated by a W-shaped pre-SFT trajectory: the procedure hurts very small bases (compounding errors) and competent-but-not-frontier bases (overhead without signal), helps an intermediate base (scaffolding skipped checks), and helps a frontier base modestly (native procedural capacity). SFT's distinctive value is in lifting Δ in the negative regimes — rescuing 0.8B from procedure-incompetent to procedure-neutral, flipping 4B from procedure-overhead to procedure-helps. The mechanism is regime-asymmetric rather than capacity-conditional, with a falsifiable prediction at 8B/14B. v2.0 (4B+SFT) ties the Haiku-4-5 frontier reference at the bench's effective ceiling under fair scoring.

The methodological observations alongside the empirical result are: a bench format-compliance artifact biases SFT-lift claims absent a format-tolerant base-model control; the artifact's remediation (LLM-only re-judge) reveals the deterministic judge had been systematically biased *against* the curated condition by 2–4.5 pp; cross-family judge cross-validation (GPT-5.4, 2800 paired episodes) bounds the judging-stage component of the corpus-author-overlap concern. Two earlier framings of this work — "format-only learning at 0.8B" (a deterministic-vs-Ollama path mismatch; Appendix A) and "SFT contribution shrinks at 4B" (the pre-SFT 4B control had not yet been measured) — were path-mismatch artifacts and are explicitly superseded. The remaining structurally unaddressed concern is corpus-generation overlap; future work should extend the cross-family bound to a third family and replicate across seeds, both inexpensive.

Acknowledgments

Compute provided by vast.ai (Qwen3.5-2B and Qwen3.5-4B training and partial evaluation) and a local consumer GPU (RTX 3080 Ti, evaluation and probing). Skill catalog content seeded from Skill-Mix [1] Tables 5 and 6 and expanded by hand. Opus 4.7 (1M context) was used for skill procedural rewriting, task synthesis, baseline trace generation, and judge-of-record scoring; the resulting judge-overlap is itemised as a threat to validity in §10.

Code and Data Availability

Repository: `https://github.com/izzortsi/skillmix-n-skillsbench` (branch `dev`). Episode-level eval logs at `data/pipeline-runs/default/bench-eval-*`. Engineering reports in `b1.reports/` (cited in Appendix D); dated, self-contained snapshots written contemporaneously with each result, included for reproducibility audit rather than as primary evidence.

References

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[9] A. Yang, A. Li, B. Yang, et al. *Qwen3 Technical Report.* arXiv:2505.09388, 2025.

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A Path-mismatch resolution for pre-SFT 0.8B

The original pre-SFT 0.8B baseline (BASELINE 0.510 / CURATED 0.565 / Δ+0.055) was run via Ollama, not the same HuggingFace transformers path used for 2B/4B. Quantization defaults, chat-template handling, and decoding-stack details differ between the two paths. The mismatch was material: under matched HF scoring, pre-SFT 0.8B lands at BASELINE 0.625 / CURATED 0.510 / $\Delta - 0.115$ (deterministic-mixed) or BASELINE 0.665 / CURATED 0.590 / $\Delta - 0.075$ (LLM-only). The Ollama-path $\Delta + 0.055$ sat on the wrong side of zero relative to both matched-path measurements, and the v1 attribution flipped accordingly.

A.1 Methodology

We resolved the mismatch in two parts:

LLM-only path. A fresh HF eval of pre-SFT 0.8B under `-force-llm-judge` (Table 2) lands at $0.665/0.590/-0.075$.

Deterministic-mixed path (Table 5's native dispatch). The 334 deterministic-type episodes from the same HF run were re-scored locally via the deterministic extractor; the 33 `FREE_FORM` tasks (66 episodes) carry their LLM-judged scores from the `-force-llm-judge` run, since under `-force-llm-judge` `FREE_FORM` tasks land on the same LLM/FREE_FORM verifier path that Table 5's deterministic-mixed dispatch uses for `FREE_FORM`. Result: 0.625/0.510/−0.115. Reproducer: `training/rejudge_failed_episodes.py -judge deterministic -all`. This is Table 5's pre-SFT 0.8B row; the original Ollama numbers are no longer presented in any headline table.

A.2 Consequence for v1 attribution

Under matched HF scoring, the v1 SFT Δ-lift is +0.065 (deterministic-mixed) or +0.070 (LLM-only) — consistent across both judging paths within 1 pp and not the −0.060 "Δ-flipped negative" the path-mismatched comparison originally suggested. The earlier "format-only learning at 0.8B" framing compared v1's deterministic $\Delta = -0.050$ against Ollama-path pre-SFT $\Delta = +0.055$, both deterministic-mixed but on different generation paths. Under the corrected matched-path pre-SFT row, SFT delivers +1.0 pp BASELINE lift, +7.5 pp CURATED lift, and +0.065 Δ-lift attributable to SFT at 0.8B — a clean differential lift rather than a Δ-flip. The 2B and 4B SFT contributions are unaffected since their endpoints were already on a single path each.

B Pipeline file structure

```
skillmix-n-skillsbench/
|-- core/                                shared schema + provider abstractions
|-- harness/                             LinearAgent: streaming + thinking-block capture
|-- b2_benchmarks/skillsbench/           corpus_harness, llm_judge, procedural_catalog
|-- s1_extracting_skill_names/           declarative-skill seeders (Wikipedia-style)
|-- s2_extracting_skills_from_text/      procedural-rewrite stage
|-- s3_generating_skill_examples/m5_*.py task synthesis from procedural skills
|-- s4_extracting_skill_usage_instances/ solver + trace + judge composition (m4)
|-- training/
|   |-- train_qwen35_lora.py             LoRA / partial-FT trainer (TRL >=0.18)
|   |-- eval_qwen35_lora.py              holdout evaluator (--adapter optional)
|   |-- patch_qwen35_template.py         chat-template generation-marker patcher
|   |-- chat_with_adapter.py             interactive + probe-set generality tester
|   |-- rejudge_failed_episodes.py       post-hoc rejudge for transient judge errors
|   `-- compare_pre_post.py              per-skill diff utility
|-- docker/                              vast.ai training image + setup docs
|-- data/pipeline-runs/default/          full run artifacts
`-- b1.reports/                          engineering log (yymmdd.title.txt)
```

C Per-skill v2.0 result (n=5 per skill, sorted by Δ)

Lift cluster (procedure helped, deterministic judging):

- accident-fallacy +0.40, spatial-reasoning +0.40
- false-dilemma, confirmation-bias, anchoring-bias, appeal-to-emotion, base-rate-neglect, transitive-inference, necessary-and-sufficient-conditions, metaphor, emotional-self-regulation: +0.20 each

Flat cluster ($\Delta = 0$): 25 skills, of which 22 are at BASELINE ≥ 0.80 (ceiling-bound).

Regression cluster (procedure hurt; all 5/5 BASELINE $\rightarrow$ 4/5 CURATED): complex-question, hypothetical-syllogism, framing-effect, equivocation. The regression cluster overlaps with deductive-logic skills where the base 4B has ceiling competence; this matches the per-skill miniature of the cross-scale regime-asymmetric finding (§3.4). Under LLM-only judging the regression cluster shrinks to one skill (equivocation), consistent with the curated-condition deterministic-judge bias (§4).

D Engineering reports cited

The repository's `b1.reports/` directory contains dated, self-contained engineering snapshots written contemporaneously with each result. These are included for reproducibility audit rather than as primary evidence; the primary evidence in this paper is the episode-level logs and the recipe-by-recipe pass rates in Tables 2 and 5.

- `260426.findings-pre-post-sft-iterations.txt` — v1–v1.8 consolidation
- `260428.sft-v1_9-2b-result.txt` — v1.9 result + attribution
- `260428.v1_9-generality-probe.txt` — v1.9 generality regression
- `260428.sft-v2_0-4b-result.txt` — v2.0 result + attribution caveat
- `260428.v2_0-generality-probe.txt` — v2.0 generality preservation
- `260506.bench-format-sensitivity-finding.txt` — pre-SFT 4B format artifact
- `260506.bench-statistical-significance.txt` — McNemar + bootstrap tests under deterministic judging
- `260506.llm-only-rejudge-findings.txt` — LLM-only re-judge across all seven configurations; the pre-SFT 4B finding ($\Delta \leq 0$), the pre-SFT 0.8B HF finding ($\Delta - 0.075$, deepest trough), and the mechanism-uniformity reframe all originate here
- `260508.cross-family-judge-validation.txt` — GPT-5.4 vs Opus 4.7 cross-family validation, 7 datasets / 2800 paired episodes, headline numbers in Table 4

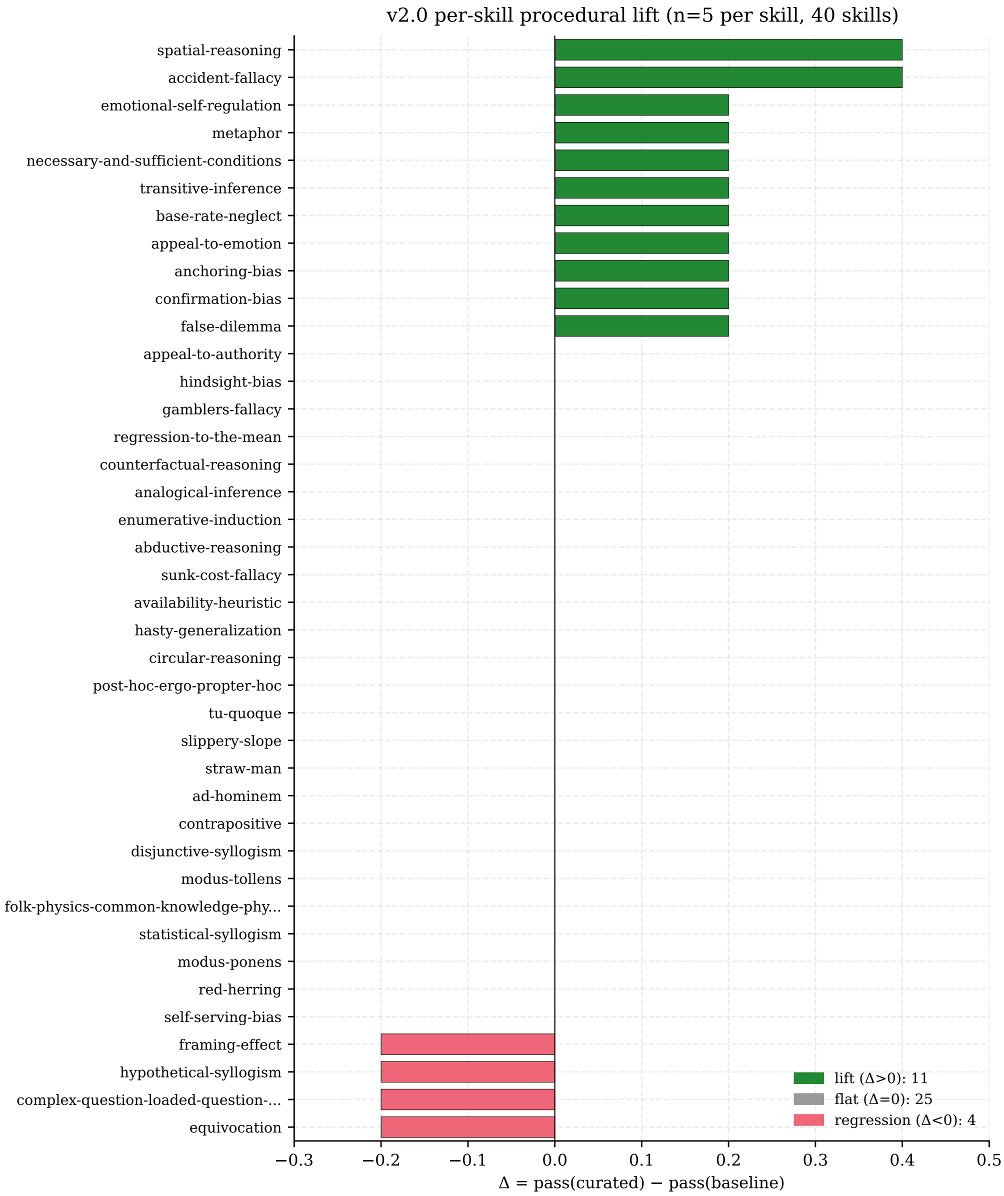

Figure 2: Per-skill procedural-Δ at v2.0 (n=5 per skill, 40 skills). 11 skills lift on CURATED; 25 are flat (mostly because BASELINE is already at ceiling); 4 regress. The lift cluster concentrates on interpretive and spatial-social skills; the regression cluster on deductive-logic skills where the model has ceiling competence on BASELINE.

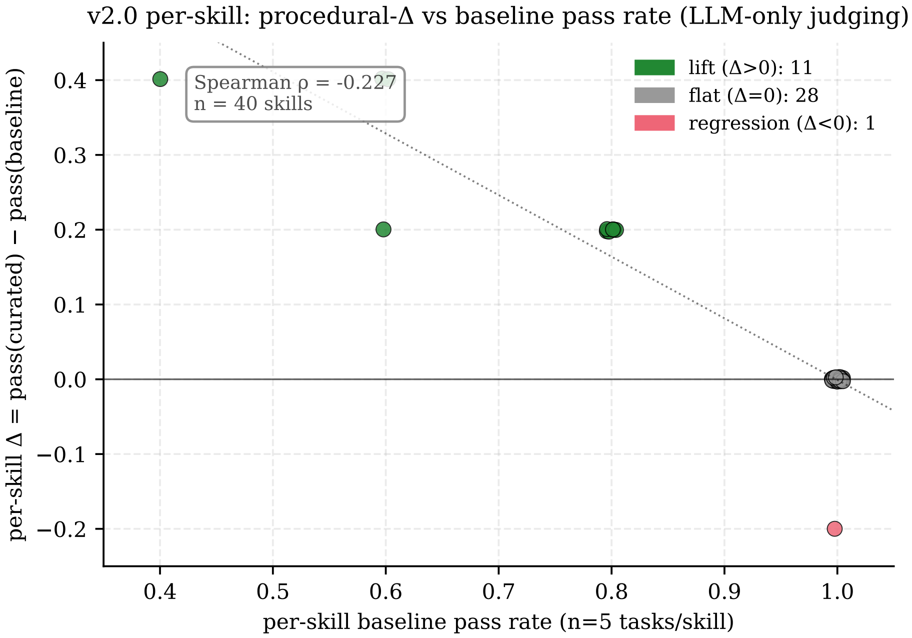

Figure 3: Per-skill procedural-Δ at v2.0 versus per-skill baseline pass rate (LLM-only judging, n=5 tasks per skill, 40 skills). Spearman $\rho = -0.227$. Skills where the base is already at ceiling on baseline cannot benefit from procedure injection (flat cluster); skills with lower baseline have the most Δ headroom (lift cluster). The negative slope is qualitative evidence that SFT's distinctive value is in scaffolding skills the base struggles with, not in lifting already-saturated skills.

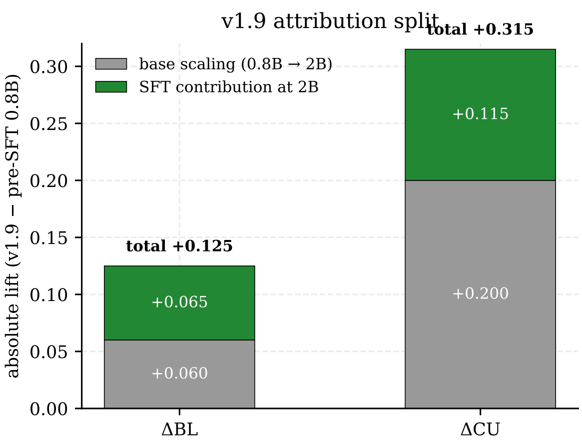

Figure 4: v1.9 lift over pre-SFT 0.8B decomposed into base-model scaling (gray; measured between the two pre-SFT controls) and SFT contribution at 2B (green; the residual from pre-SFT 2B to v1.9). The SFT contribution is +6.5 pp on BASELINE and +11.5 pp on CURATED.

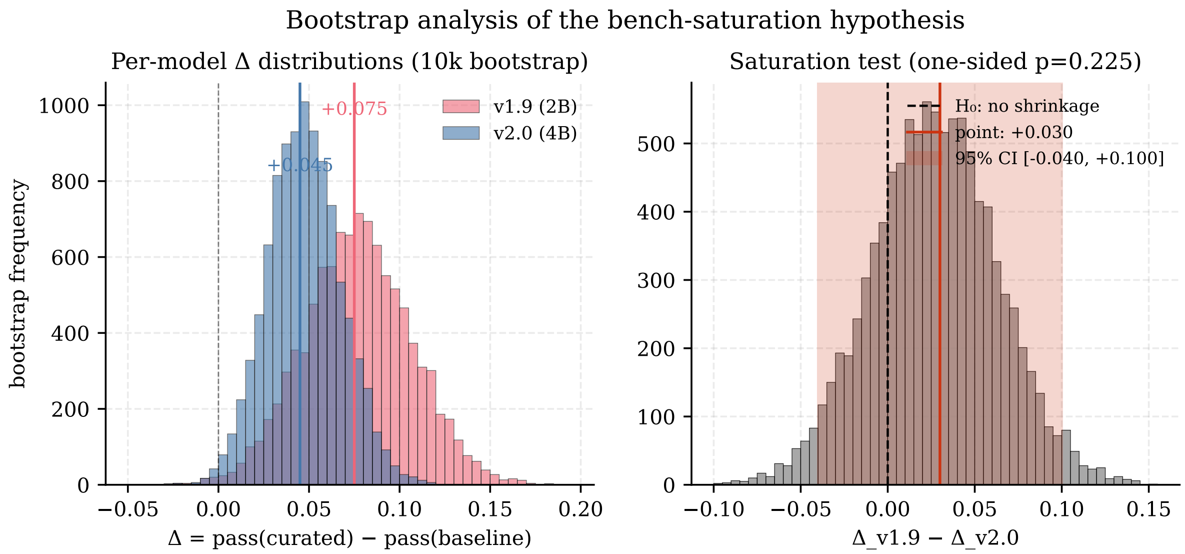

Figure 5: Left: bootstrap distributions of the procedural-Δ for v1.9 and v2.0 (10,000 resamples of paired `task_uids`). Both distributions sit above zero (dashed line) but overlap substantially. Right: distribution of $\Delta_{\mathrm{v1.9}} - \Delta_{\mathrm{v2.0}}$; the 95% CI (red shading) includes zero, and the one-sided test for the saturation hypothesis returns p=0.225. Direction consistent with saturation; magnitude not statistically distinguishable from noise.